

SELECTION OF GOOD POPULATIONS: THE SCALE PARAMETER CASE

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Summary: Let $\Pi_1, \dots, \Pi_k$ be $k(\geq 2)$ independent populations such that observations from population Π_i have an absolutely continuous distribution function $F_{\theta_i}(x) = F\left[\frac{x}{\theta_i}\right]$, indexed by the scale parameter $\theta_i > 0, i = 1, \dots, k$. Let $\theta_{(i)}$ denote the i -th smallest of $\theta_1, \dots, \theta_k$. The population Π_i is defined to be 'good' if $\theta_i \geq \delta_1 \theta_{(k)}, i = 1, \dots, k$, where $\delta_1 \in (0, 1)$. We propose a subset selection procedure which includes all the good populations in the selected subset with probability at least P^* , a pre-specified value in $(0, 1)$. Simultaneous inference using the proposed selection procedure is discussed. If the populations with scale parameter less than $\delta_2 \theta_{(k)}$ are 'bad', where $\delta_2 \in (0, 1)$ and $\delta_2 < \delta_1$, a selection procedure is proposed so that the probability of either selecting a bad population or omitting a good population is at most $1 - P^*$.

1. INTRODUCTION

Let $\Pi_1, \dots, \Pi_k$ be $k(\geq 2)$ independent populations such that observations from population Π_i have an absolutely continuous distribution function

$F_{\theta_i}(x) = F(\frac{x}{\theta_i})$, indexed by the scale parameter $\theta_i > 0, i = 1, \dots, k$. Let X_{ij} ($i = 1, \dots, k; j = 1, \dots, n$) be independent random samples from the k populations. Let $Y_1, \dots, Y_k$ be independent random variables (r.v.'s) which characterize the populations $\Pi_1, \dots, \Pi_k$ respectively (these r.v.'s will generally be consistent estimators of the θ 's based on samples of common size n from each population). Let the ordered values of unknown θ 's be denoted by $\theta_{(1)} \leq \dots \leq \theta_{(k)}$. In case of ties, the indices can be chosen in any manner so long as the above order relation is preserved. Let $Y_{[i]}$ denote the i -th smallest of $Y_1, \dots, Y_k$. For a given $\delta_i \in (0, 1), i = 1, 2$, the population Π_i is said to be 'good' if $\theta_i \geq \delta_1 \theta_{(k)}$ and 'bad' if $\theta_i < \delta_2 \theta_{(k)}$, where $\delta_2 < \delta_1$. The goal is to select a subset $S \subseteq \{1, \dots, k\}$ which contains all the good populations with probability P^* , a prespecified value.

Desu (1970) restricted the criterion of 'badness' and defined a set of bad populations as $B = \{i : \theta_i < \delta_2 \theta_{(k)}\}$. He proposed a subset selection procedure for which the correct selection is achieved if $S \subseteq B^c$, where B^c is the complement of set B . In the present paper we consider a selection goal dual to that considered by Desu (1970). Let $G = \{i : \theta_i \geq \delta_1 \theta_{(k)}\}$ be the set of all good populations. Following Lam's (1986) approach we propose a subset selection procedure to select a subset which contains all the populations in G with probability at least P^* .

In section 2 of this paper we have proposed the selection procedure followed by applications to exponential and normal distributions. Simultaneous confidence intervals for the ratios $\frac{\theta_{(k)}}{\theta_i}, i = 1, \dots, k$ and $\frac{\theta_{(i)}}{\theta_{(j)}}, i \neq j, i, j = 1, \dots, k$, that can be derived with the help of the proposed selection procedure, are given in section 3. A selection procedure is proposed in section 4 under which the probability of either omitting a good population or selecting a bad population is less than or equal to $1 - p^*$.

2. PROPOSED SELECTION PROCEDURE

Let $G = \{i : \theta_i \geq \delta \theta_{(k)}\}$, be the set of all good populations where $\delta \in (0, 1)$ is a given constant. We desire to select a random-sized subset S of k populations which includes the set G . The selection of any subset S such that $G \subseteq S$ is called a correct selection (CS). For a given $P^* \in (0, 1)$, a subset selection procedure R is said to satisfy P^* -condition if

$$(2.1) \quad P_{\underline{\theta}}[CS|R] \geq P^*$$

for all $\theta_1, \dots, \theta_k$ in the parameter space $\Omega = \{\underline{\theta} : \underline{\theta} = (\theta_1, \dots, \theta_k)^t, \text{ all } \theta_i > 0\}$.

Let $q = q(k; P^*, n)$ be the P^* quantile of the distribution of random variable Z defined as

$$(2.2) \quad Z = \frac{\max_{1 \leq i \leq k} \left[\frac{Y_i}{\theta_i} \right]}{\min_{1 \leq i \leq k} \left[\frac{Y_i}{\theta_i} \right]}$$

The proposed selection procedure is

R : Include the population Π_i in the subset S if and only if

$$Y_i \geq \frac{\delta}{q} Y_{[k]}$$

That is, $S = \{i : Y_i \geq \frac{\delta}{q} Y_{[k]}\}$

The following theorem shows that the selection procedure R satisfies the probability requirement (2.1).

THEOREM 2.1. Let $G = \{i : \theta_i \geq \delta \theta_{(k)}\}$ and $q = q(k; P^*, n)$ be the P^* quantile of the distribution of random variable Z defined in (2.2). Then

$$(2.3) \quad P_{\underline{\theta}}[G \subseteq S] \geq P^* \quad \text{for all } \underline{\theta} \in \Omega$$

PROOF: Define

$$(2.4) \quad T = \{Z \leq q\}.$$

Let $\theta_{[i]}$ denote the parameter associated with $Y_{[i]}$, $i = 1, \dots, k$. Then

$$\begin{aligned} T &= \left\{ \max_{1 \leq i \leq k} \left[\frac{Y_i}{\theta_i} \right] \leq q, \min_{1 \leq i \leq k} \left[\frac{Y_i}{\theta_i} \right] \right. \\ &\quad \left. \leq \min_{1 \leq i \leq k} \left[\frac{Y_i}{\theta_i} \right] \leq \frac{1}{q} \frac{Y_{[k]}}{\theta_{[k]}} \right\} \\ &= \left\{ \min_{1 \leq i \leq k} \left[\frac{Y_i}{\theta_i} \right] \geq \frac{1}{q} \frac{Y_{[k]}}{\theta_{[k]}} \right\} \\ &= \left\{ Y_i \geq \frac{1}{q} \frac{\theta_i}{\theta_{(k)}} Y_{[k]}, i = 1, \dots, k \right\} \end{aligned}$$

Thus T and G imply the event

$$\left\{ Y_i \geq \frac{\delta_i}{q} Y_{[k]}, i = 1, \dots, k \right\} = S.$$

Therefore,

$$P^* = P_{\theta}(T) \leq P_{\theta}(G \subseteq S), \forall \theta \in \Omega.$$

2.1 Application to the largest exponential scale parameter

Let $X_{i1}, \dots, X_{in}$ be the random sample taken from the exponential population Π_i , which has location parameter μ_i and scale parameter θ_i . Let $X_i = \min_{1 \leq j \leq n} X_{ij}$ denote the smallest order statistic of the sample from Π_i . The minimum variance unbiased estimate of θ_i is

$$Y_i = \frac{1}{n-1} \sum_{j=1}^n (X_{ij} - X_i)$$

and Y_i has a gamma distribution with parameters $n-1$ and $\frac{\theta_i}{n-1}$. (3.3)

For selecting a subset which contains all the populations with $\theta_i \geq \delta_i$, we use the selection procedure R . The selection constant q is the solution of the equation

$$P[Z \leq q] = P^*$$

with $Z = \frac{\max_{1 \leq i \leq k} (Y_i/\theta_i)}{\min_{1 \leq i \leq k} (Y_i/\theta_i)}$, the extremal quotient of iid random variables Y_i .

Given k, n, P^* the value of q can be obtained from the table for Hartley's ratio Z with $2(n-1)$ and $2(n-1)$ degrees of freedom (Pearson & Hartley (1970)).

2.2 Application to the largest normal variance

Let $X_{11}, \dots, X_{in}$ be a random sample of size n from the normal population Π_i with mean μ_i and variance $\theta_i = \sigma_i^2$, $i = 1, \dots, k$. Then the uniformly minimum variance unbiased estimator of θ_i is

$$Y_i = S_i^2 = \frac{1}{n-1} \sum_{j=1}^n (X_{ij} - \bar{X}_i)^2$$

and $(n-1)Y_i/\theta_i$ has a chi-square distribution with $(n-1)$ degrees of freedom. Now for a given $\delta \in (0, 1)$, we can use selection procedure R to select a subset which includes all the populations with $\theta_i \geq \delta \theta_{(k)}$. The selection constant $q(>1)$ is obtained from the equation

$$P[Z \leq q] = P^*$$

where $Z = \frac{\max_{1 \leq i \leq k} (Y_i/\theta_i)}{\min_{1 \leq i \leq k} (Y_i/\theta_i)}$, the extremal quotient of iid random variables

Y_i . The value of q can be obtained from the table in Pearson and Hartley (1970) with $2(n-1)$ and $2(n-1)$ degrees of freedom for the equality of normal variances.

In the following section we discuss the simultaneous inference that can be derived from the proposed selection procedure.

3. Simultaneous confidence intervals for all ratios to the best

Multiple comparison with the best population was first proposed by Hsu (1981), Edwards and Hsu (1983), Hsu (1984) followed by Lam (1986), Chen and Vanichbuncha (1989), and Lam (1989). In all these cases the authors have restricted themselves either to parametric or non-parametric location models and proposed simultaneous confidence intervals for the distances measured by taking the difference between the location parameters of interest. Likewise under the scale model one may take the deviation between the two populations in terms of the ratio of their scale parameters and may consider proposing simultaneous confidence intervals for the deviations measured by taking the ratio between the scale parameters of interest. For the scale model considered here, simultaneous confidence intervals for the parameters $\alpha_i = \frac{\theta_i}{\theta_1}$, $i = 1, \dots, k$ and for the ratios $\frac{\theta_i}{\theta_j}$, $i \neq j$, of ranked

scale parameters are derived, using the proposed selection procedure, without decreasing the nominal confidence level P^* . The results are summarized in the following two theorems.

THEOREM 3.1. Under the assumptions of theorem 2.1, we have

$$P \left[G \subseteq S, \max \left[\frac{1}{q} \frac{Y_{[k]}}{Y_i}, 1 \right] \leq \frac{\theta_{(k)}}{\theta_i} \leq \max \left[q \left[\max_{j \neq i} \frac{Y_j}{Y_i} \right], 1 \right], i = 1, \dots, k \right] \geq P^*.$$

PROOF: Define T_i as in (2.4) and let $Y_{(k)}$ denote the random variable associated with $\theta_{(k)}$. Then as in the proof of theorem 2.1 we have

$$(3.1) \quad T \subseteq \left\{ \frac{\theta_{(k)}}{\theta_i} \geq \frac{1}{q} \frac{Y_{[k]}}{Y_i}, i = 1, \dots, k \right\} \\ = \left\{ \frac{\theta_{(k)}}{\theta_i} \geq \max \left[\frac{1}{q} \frac{Y_{[k]}}{Y_i}, 1 \right], i = 1, \dots, k \right\}.$$

Moreover,

$$\begin{aligned}
 (3.2) \quad T &= \left\{ \frac{T_i}{\theta_i} \leq q \min_{i \leq i \leq k} \left[\frac{Y_i}{\theta_i} \right], i = 1, \dots, k \right\} \\
 &\subset \left\{ \frac{Y_i}{\theta_i} \leq q \frac{Y_{(k)}}{\theta_{(k)}}, i = 1, \dots, k \right\} \\
 &= \left\{ \frac{\theta_{(k)}}{\theta_i} \leq q \frac{Y_{(k)}}{Y_i}, i = 1, \dots, k \right\} \\
 &\subset \left\{ \frac{\theta_{(k)}}{\theta_i} \leq q \max_{j \neq i} \frac{Y_j}{Y_i}, i = 1, \dots, k \right\} \\
 &= \left\{ \frac{\theta_{(k)}}{\theta_i} \leq \max \left[q \max_{j \neq i} \frac{Y_j}{Y_i}, 1 \right], i = 1, \dots, k \right\}.
 \end{aligned}$$

Furthermore as in the proof of theorem 2.1

$$(3.3) \quad T \subset \{G \subseteq S\}.$$

Thus from (3.1), (3.2) and (3.3) we have

$$P^* = P_{\theta}(T) \leq P \left[G \subseteq S, \max \left[\frac{1}{q} \frac{Y_{(k)}}{Y_i}, 1 \right] \leq \frac{\theta_{(k)}}{\theta_i} \leq \max \left[q \left[\max_{j \neq i} \frac{Y_j}{Y_i} \right], 1 \right], i=1, \dots, k \right]$$

and the result follows.

Below we state two lemmas which will be used to derive simultaneous

confidence intervals for the ratios $\frac{\theta_{(i)}}{\theta_{(j)}}, i \neq j$, of ranked θ -values.

LEMMA 3.1: For any i between 1 and k ,

$$\min_{j \leq i} \frac{Y_{[j]}}{\theta_{[j]}} \leq \frac{Y_{[i]}}{\theta_{(i)}} \leq \max_{j \geq i} \frac{Y_{[j]}}{\theta_{[j]}}.$$

PROOF:

$$(3.4) \quad \min_{j \leq i} \frac{Y_{[j]}}{\theta_{[j]}} \leq \min_{j \leq i} \frac{Y_{[j]}}{\theta_{[j]}} \leq \frac{Y_{[i]}}{\theta_{(i)}}.$$

Similarly, we can show that

$$(3.5) \quad \frac{Y_{[i]}}{\theta_{(i)}} \leq \max_{j \geq i} \frac{Y_{[j]}}{\theta_{[j]}}.$$

Thus the lemma follows from (3.4) and (3.5).

LEMMA 3.2: We have the inequalities

$$\min_{1 \leq i \leq k} \frac{Y_{[i]}}{\theta_{[i]}} \leq \min_{1 \leq i \leq k} \frac{Y_{[i]}}{\theta_{(i)}}, \quad \max_{1 \leq i \leq k} \frac{Y_{[i]}}{\theta_{(i)}} \leq \max_{1 \leq i \leq k} \frac{Y_{[i]}}{\theta_{[i]}}.$$

The proof of this lemma follows from lemma 3.1 and is hence omitted.

THEOREM 3.2: Assume that the assumptions of theorem 2.1 hold. Then

$$P \left[\frac{Y_{[j]}}{q Y_{[j]}} \leq \frac{\theta_{(i)}}{\theta_{(j)}} \leq q \frac{Y_{[i]}}{Y_{[j]}}, \quad i, j = 1, \dots, k \right] \geq P^*.$$

PROOF: Recall the definition of T in (2.4) and from lemma 3.2, we have

$$(3.6) \quad \begin{aligned} T &= \left\{ \max_{1 \leq j \leq k} \frac{Y_{[j]}}{\theta_{[j]}} \leq q \min_{1 \leq i \leq k} \frac{Y_{[i]}}{\theta_{[i]}} \right\} \\ &\subset \left\{ \max_{1 \leq j \leq k} \frac{Y_{[j]}}{\theta_{(j)}} \leq q \min_{1 \leq i \leq k} \frac{Y_{[i]}}{\theta_{(i)}} \right\} \\ &= \left\{ \frac{\theta_{(i)}}{\theta_{(j)}} \leq q \frac{Y_{[i]}}{Y_{[j]}}, \quad i, j = 1, \dots, k \right\}. \end{aligned}$$

Also from lemma 3.2, we have

$$(3.7) \quad \begin{aligned} T &= \left\{ \min_{1 \leq j \leq k} \frac{Y_{[j]}}{\theta_{[j]}} \geq \frac{1}{q} \max_{1 \leq i \leq k} \frac{Y_{[i]}}{\theta_{[i]}} \right\} \\ &\subset \left\{ \min_{1 \leq j \leq k} \frac{Y_{[j]}}{\theta_{(j)}} \geq \frac{1}{q} \max_{1 \leq i \leq k} \frac{Y_{[i]}}{\theta_{(i)}} \right\} \\ &= \left\{ \frac{\theta_{(i)}}{\theta_{(j)}} \geq \frac{1}{q} \frac{Y_{[i]}}{Y_{[j]}}, \quad i, j = 1, \dots, k \right\}. \end{aligned}$$

On combining (3.6) and (3.7), we get

$$T \subset \left\{ \frac{Y_{[j]}}{qY_{[j]}} \leq \frac{\theta_{(i)}}{\theta_{(j)}} \leq q \frac{Y_{[j]}}{Y_{[j]}}, i, j = 1, \dots, k \right\}.$$

Therefore

$$P^* = P_{\theta}(T) \leq P \left[\frac{Y_{[j]}}{qY_{[j]}} \leq \frac{\theta_{(i)}}{\theta_{(j)}} \leq q \frac{Y_{[j]}}{Y_{[j]}}, i, j = 1, \dots, k \right].$$

In the following section we propose a subset selection procedure which controls the probability of both types of errors.

4. CONTROLLING BOTH TYPES OF ERRORS

In proposing a selection procedure an experimenter is faced with two types of errors. One type of error is to omit a good population and the other is to include a bad population. Let an experimenter classify a population Π_i as 'good' if $\theta_i \geq \delta_1 \theta_{(k)}$, $\delta_1 \in (0, 1)$, and classify a population Π_i as 'bad' if $\theta_i < \delta_2 \theta_{(k)}$, where $\delta_2 < \delta_1$ and $\delta_2 \in (0, 1)$. Let

$$G = \{i : \theta_i \geq \delta_1 \theta_{(k)}\}, B = \{i : \theta_i < \delta_2 \theta_{(k)}\}.$$

Then the goal of the experimenter is to control both types of errors and furthermore he wants the selected subset to satisfy $G \subseteq S \subseteq B^c$ with high probability, where B^c is the complement of set B .

In the following theorem we propose a selection procedure which meets the above goal with probability greater than or equal to P^* .

THEOREM 4.1: Let $\Pi_1, \dots, \Pi_k$ be k (≥ 2) independent populations such that the population Π_i has an absolutely continuous distribution function $F_{\theta_i}(x) = F\left[\frac{x}{\theta_i}\right]$, $i = 1, \dots, k$. Let $G = \{i : \theta_i \geq \delta_1 \theta_{(k)}\}$ and $B^c = \{i : \theta_i \geq \delta_2 \theta_{(k)}\}$, where $\delta_1 \in (0, 1)$, $i = 1, 2$ and $\delta_2 < \delta_1$. Let δ_1 and δ_2 be chosen such that $\frac{\delta_1}{\delta_2} > q^2 = q^2(k; P^*; n)$ and let t be a constant satisfying $\frac{\delta_1}{q} > t > \delta_2 q$. Choose $S = \{i : Y_i \geq t Y_{[k]}\}$. Then

$$P[G \subseteq S \subseteq B^c] \geq P^*.$$

PROOF: Let T be defined as in (2.4). Then

$$(4.1) \quad T \subseteq \left\{ \frac{\max_{1 \leq i \leq k} \left[\frac{Y_i}{\theta_i} \right]}{\min_{1 \leq i \leq k} \left[\frac{Y_i}{\theta_i} \right]} \leq \frac{\delta_1}{t} \right\}$$

Following the lines of the proof of theorem 2.1, it can easily be shown that the event in the right-hand side of (4.1) is contained in the event $(G \subseteq S)$.

Therefore,

$$(4.2) \quad T \subseteq (G \subseteq S).$$

Also

$$(4.3) \quad [S \subseteq B^c] = [B \subseteq S^c]$$

$$= [Y_i < tY_{[k]} \text{ for all } i \text{'s s.t. } \theta_i < \delta_2 \theta_{(k)}]$$

$$\supset \left[\frac{Y_{(k)}}{Y_i} > \frac{1}{t} \text{ for all } i \text{'s s.t. } \theta_i < \delta_2 \theta_{(k)} \right]$$

$$= \left[\frac{Y_{(k)}/\theta_{(k)}}{Y_i/\theta_i} > \frac{1}{t} \frac{\theta_i}{\theta_{(k)}} \text{ for all } i \text{'s s.t. } \theta_i < \delta_2 \theta_{(k)} \right]$$

$$\supset \left[\frac{Y_{(k)}/\theta_{(k)}}{Y_i/\theta_i} > \frac{\delta_2}{t} \text{ for all } i \text{'s s.t. } \theta_i < \delta_2 \theta_{(k)} \right]$$

$$\supset \left[\frac{Y_{(k)}/\theta_{(k)}}{Y_i/\theta_i} > \frac{\delta_2}{t}, i = 1, \dots, k \right]$$

$$\supset \left[\frac{Y_{(k)}/\theta_{(k)}}{Y_i/\theta_i} > \frac{1}{q}, i = 1, \dots, k \right]$$

$$= \left[\frac{Y_i/\theta_i}{Y_{(k)}/\theta_{(k)}} < q, i = 1, \dots, k \right]$$

$$\supset T.$$

On combining (4.2) and (4.3), we get

$$T \subset [G \subset S \subset B^c].$$

Therefore,

$$P^* = P(T) \leq P_{\theta}(G \subset S \subset B^c) \quad \forall \theta \in \Omega.$$

Hence the result follows.

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